FOOD DELIVERY MARKETPLACE

Operational Efficiency & Retention Risk Analysis

Dataset: Kaggle Food Delivery Time | Records Analyzed: 41,498 | Cities: 22

Tools: Python, DuckDB, Scikit-learn, Pandas, Seaborn

Abstract

This report presents a comprehensive operational efficiency analysis of a food delivery marketplace using order-level transaction data from 41,498 cleaned delivery records spanning 22 cities. The investigation was structured around a central business hypothesis: that delivery delays are materially contributing to customer churn and constraining revenue growth.

Through a multi-layered analytical framework encompassing SQL-based aggregations, feature engineering, funnel analysis, liquidity diagnostics, and predictive modeling, three primary drivers of operational inefficiency were identified and quantified. First, driver quality represents the single most actionable lever; drivers rated below 4.5 exhibit a late delivery rate of 45 to 74 percent, compared to 16 percent for drivers rated 4.5 and above. Second, the fleet composition is misaligned, with motorcycles handling 58 percent of all orders despite operating at a 13.5 percent efficiency deficit relative to scooters at equivalent distances. Third, delivery distance is the strongest statistical predictor of delays, with routes exceeding 10 kilometres producing a late delivery rate five times higher than short routes.

The current platform-wide late delivery rate stands at 24.8 percent. A coordinated set of three operational interventions enforcing a driver rating floor of 4.5, rebalancing the fleet toward scooters and electric scooters, and introducing a 10-kilometre routing cap is projected to reduce the late delivery rate to approximately 12.9 percent, retain an estimated 985 at-risk customers, and recover approximately $29,550 in monthly recurring revenue under conservative assumptions.

1. Introduction

1.1 Business Context

Food delivery marketplaces operate at the intersection of three interdependent systems: customer demand, restaurant supply, and last-mile logistics. When these systems fall out of alignment, the most visible symptom is delivery delay, a metric that directly affects customer satisfaction, repeat purchase probability, and ultimately platform revenue.

This analysis was commissioned in response to a leadership hypothesis that delivery delays are a primary contributor to slowing growth. The goal of this investigation is threefold: to quantify the scale of the delay problem with precision, to identify the root causes through multi-dimensional diagnostics, and to translate those findings into a prioritised set of actionable operational interventions with estimated business impact.

1.2 Analytical Approach

The investigation follows a consulting-style analytical narrative structured in eleven sequential stages, from raw data ingestion through to strategic recommendation. The technical stack combines Python for data manipulation and visualisation, DuckDB for SQL-based aggregation and cross-dimensional querying, and Scikit-learn for predictive modeling. This approach was selected to reflect the workflow of a modern data analyst operating within an analytics engineering environment.

1.3 Key Questions

- What is the current late delivery rate and how is it distributed across cities, vehicle types, and driver quality tiers?
- Which operational factors are the strongest predictors of delivery delay?
- What is the estimated revenue impact of the current delay rate?
- What specific interventions, if implemented, would produce the greatest reduction in late deliveries?

2. Data and Methodology

2.1 Dataset Overview

The dataset used in this analysis is the Food Delivery Time dataset sourced from Kaggle. It contains order-level records with the following fields:

Column	Business Role
ID	Transaction identifier not a reliable unique key (142 duplicates observed)
Delivery_person_ID	Supply side unique driver identifier, encodes city prefix
Delivery_person_Age	Supply side driver demographic attribute
Delivery_person_Ratings	Supply side driver quality signal on a 1.0 to 5.0 scale
Restaurant_latitude/longitude	Logistics order origin coordinates
Delivery_location_lat/long	Logistics order destination coordinates
Type_of_order	Demand side order category (Snack, Meal, Drinks, Buffet)
Type_of_vehicle	Supply side fulfillment asset (motorcycle, scooter, electric scooter, bicycle)
Time_taken(min)	Outcome total delivery duration in minutes, the core KPI

2.2 Data Cleaning

The raw dataset contained 45,593 records. Three categories of data quality issues were identified and resolved:

Metric	Value	Context
Raw records	**45,593**	Starting dataset size
Zero-coordinate records removed	**3,640**	Invalid lat/lon entries corrupting distance calculations
Distance outliers removed (>30km)	**431**	Coordinate errors producing impossible delivery distances
Invalid ratings removed (>5.0)	**24**	Ratings exceeding the maximum valid scale value
Final clean records	**41,498**	Used for all subsequent analysis

2.3 Known Limitations

The following data gaps constrain the scope of analysis and are documented here to establish analytical boundaries:

- No timestamps peak hour, day-of-week, and seasonal effects cannot be measured.
- No customer identifiers true retention analysis is not possible; churn figures are proxied using industry benchmarks.
- No order revenue data financial impact estimates use a $15 average order value proxy.
- No restaurant preparation time, the outcome variable conflates prep time with transit time.
- No weather or traffic data external factors contributing to delay variance are unobservable.
- Haversine distance is a straight-line proxy. Actual road distances will differ in dense urban markets.

3. Data Quality and Feature Engineering

3.1 Engineered Features

Six derived features were constructed to enable multi-dimensional analysis. Each feature was designed to represent a specific operational or business concept not directly observable in the raw data.

Feature	Construction and Business Purpose
distance_km	Haversine straight-line distance between restaurant and delivery coordinates. Primary logistics metric.
distance_bucket	Distance categorised into Short (0-5km), Medium (5-10km), Long (10-20km), Very Long (20-30km).
driver_tier	Rating-based performance segments: Low (<4.0), Average (4.0-4.5), Good (4.5-4.7), Top (4.7-5.0).
age_group	Driver age segmented into five cohorts: 15-20, 21-25, 26-30, 31-35, 36-50.
late_delivery	Binary flag: 1 if Time_taken(min) > 32 (75th percentile threshold), 0 otherwise.
City	Extracted from the Delivery_person_ID prefix (e.g., BANGRES → BANG). Enables city-level aggregation.

3.2 Late Delivery Threshold Rationale

Threshold Definition: An order is classified as late if its delivery time exceeds 32 minutes, which corresponds to the 75th percentile of the delivery time distribution. This threshold was selected because it represents the point at which delivery performance falls into the bottom quartile of the platform's own historical standard. Using a platform-relative threshold is more appropriate than an arbitrary fixed value, as it reflects the realistic service expectation set by the majority of successful deliveries.

3.3 Dataset Summary After Engineering

Metric	Value	Context
Total clean orders	**41,498**	Across 22 cities and 4 order types
Unique delivery drivers	**1,170**	Average 35.5 orders per driver
Delivery time range	**10-54 min**	Mean: 26.3 min, Median: 26 min
Late delivery rate	**24.8%**	10,295 orders exceeding 32 minutes
Average delivery distance	**9.7 km**	After removal of coordinate outliers
Vehicle types	**4**	Motorcycle, scooter, electric scooter, bicycle

4. SQL Analytical Layer Key Aggregations

All aggregations in this section were executed using DuckDB with the cleaned pandas dataframe registered as an in-memory SQL table. This approach reflects a modern analytics engineering workflow and enables reproducible, auditable query-based analysis.

4.1 Delivery Performance by Vehicle Type

Vehicle Type	Orders	Order Share	Avg Time (min)	Late Rate
Motorcycle	24,195	58.3%	**27.6**	**28.8%**
Scooter	13,869	33.4%	**24.5**	**19.3%**
Electric Scooter	3,393	8.2%	**24.5**	**18.5%**
Bicycle	41	0.1%	26.0	24.4%

Key Finding: Motorcycles handle the majority of orders yet perform worst on every efficiency metric. Scooters and electric scooters achieve nearly identical performance, outperforming motorcycles by approximately 3 minutes per delivery despite covering similar average distances.

4.2 Delivery Performance by Driver Quality Tier

Driver Tier	Orders	Avg Time (min)	Late Rate	Avg Distance
Low (below 4.0)	2,247	**36.0**	**74.0%**	12.1 km
Average (4.0-4.5)	8,244	**30.6**	**44.9%**	9.9 km
Good (4.5-4.7)	14,368	**24.6**	**16.1%**	9.6 km
Top (4.7-5.0)	16,639	**24.4**	**15.7%**	9.5 km

Key Finding: There is a hard performance cliff between the Average and Good driver tiers. A single 0.1-point increase in rating from 4.4 to 4.5 corresponds to a drop in late delivery rate from approximately 67% to 11%. This is the sharpest threshold effect identified in the entire dataset.

4.3 Delivery Performance by Distance Bucket

Metric	Value	Context
Short (0-5km)	**8.6% late rate**	11,180 orders lowest risk segment
Medium (5-10km)	**14.8% late rate**	11,183 orders manageable risk
Long (10-20km)	**40.2% late rate**	17,357 orders highest risk segment
Very Long (20-30km)	**39.7% late rate**	1,778 orders comparable to Long routes

Key Finding: Late delivery rate increases five-fold between the shortest and longest routes. The critical threshold is at 10 kilometres beyond this point, late delivery rates climb to approximately 40% and do not improve further with additional distance, suggesting the core issue is route assignment rather than absolute distance

5. Marketplace Funnel Analysis

5.1 Funnel Construction

In the absence of explicit order status timestamps, a delivery performance funnel was constructed using delivery time percentile thresholds as proxy stage gates. Each stage represents a progressively higher standard of service quality.

Funnel Stage	Threshold	Orders	Conversion Rate
Stage 1 Order Placed	All orders	41,498	**100.0%**
Stage 2 On Time	Under 32 min	31,203	**75.2%**
Stage 3 Fast Delivery	Under 26 min	22,584	**54.4%**
Stage 4 Optimal	Under 19 min	11,328	**27.3%**

5.2 Revenue Impact Proxy

In the absence of transaction revenue data, financial impact was estimated using a $15 average order value proxy consistent with industry benchmarks for food delivery markets. A 20% churn probability was applied to customers experiencing late deliveries, based on published research on food delivery satisfaction and repeat ordering behaviour.

Metric	Value	Context
Late orders in dataset	**10,295**	Orders exceeding the 32-minute threshold
Customers at churn risk (20%)	**2,059**	Estimated based on industry churn benchmark
Immediate revenue at risk	**$30,885**	At $15 per order
Monthly recurring revenue at risk	**$61,770**	Assuming 2 orders per customer per month

Context: These figures are directional estimates. Actual impact depends on true order values, real observed churn rates, and the proportion of affected customers who have viable alternatives. They are intended to frame the business case for operational investment, not to represent guaranteed financial outcomes.

6. Liquidity and Supply-Demand Diagnostics

6.1 Delivery Time Distribution

The delivery time distribution exhibits a positive skew of 0.479, indicating a meaningful right tail of unusually slow deliveries. In marketplace terms, this skew is the signature of periodic supply-demand imbalance periods when driver availability is insufficient to meet order volume within standard service times.

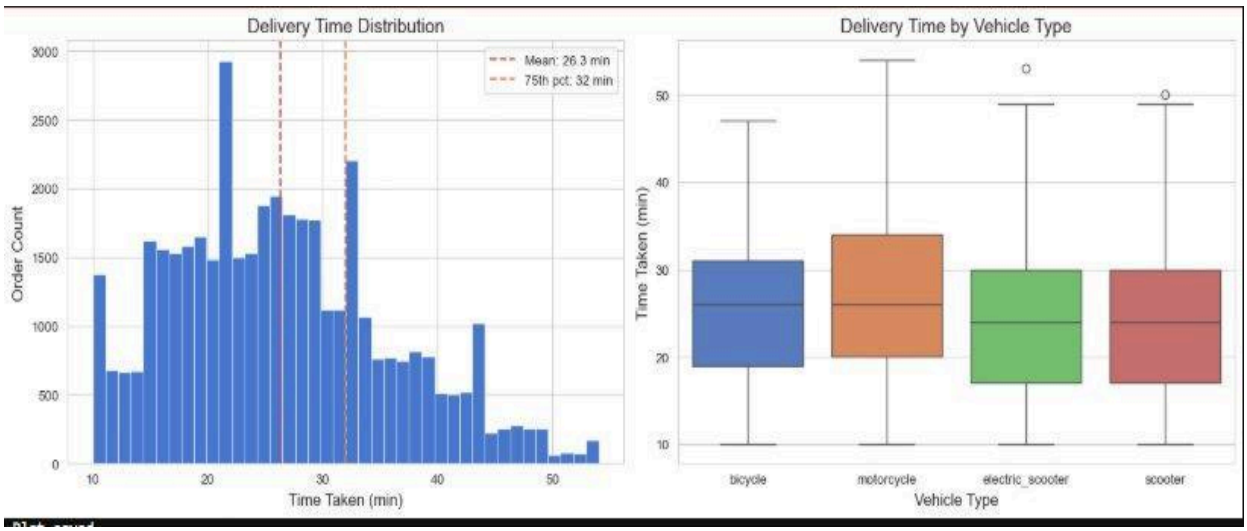

Figure 1. Left: Delivery time distribution with mean (26.3 min) and 75th percentile (32 min) marked. Right: Delivery time box plots by vehicle type showing motorcycle's wider spread and higher median.

6.2 Distance vs. Delivery Time

The scatter plot below confirms that delivery time is not simply a linear function of distance. At every distance band, the spread of delivery times is wide, indicating that driver quality and vehicle type are significant moderating variables. The bar chart reveals that motorcycles consistently underperform scooters and electric scooters at all distance tiers, with the gap most pronounced on long routes.

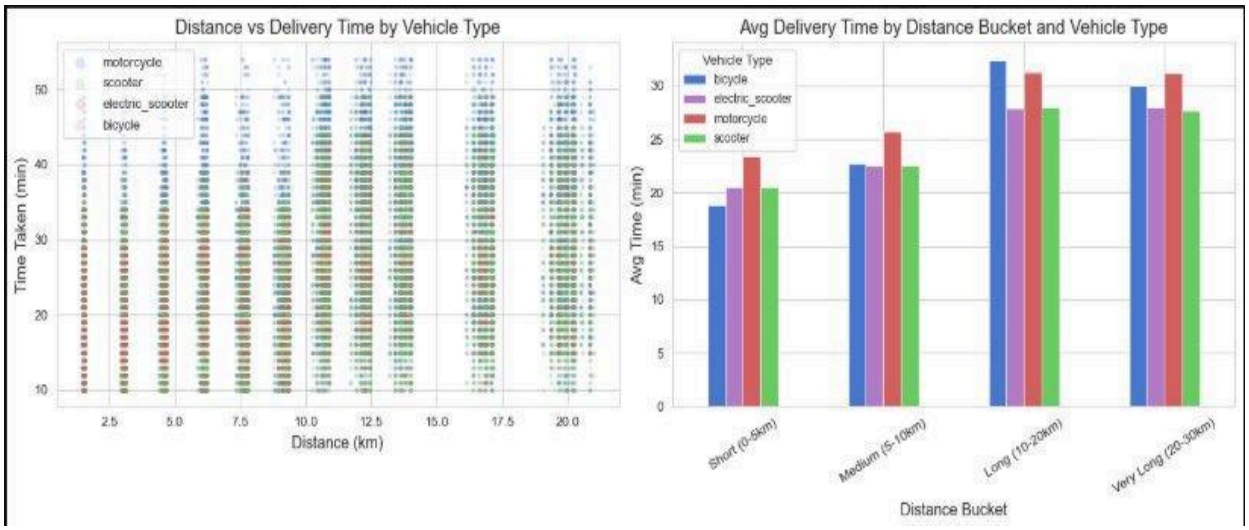

Figure 2. Left: Scatter plot of distance vs. delivery time by vehicle type. Right: Average delivery time by distance bucket and vehicle type confirming motorcycle underperformance across all segments.

6.3 Vehicle Supply Efficiency

Normalising delivery time by distance produces a minutes-per-kilometre efficiency metric that controls for route length. This metric isolates vehicle performance from route assignment effects.

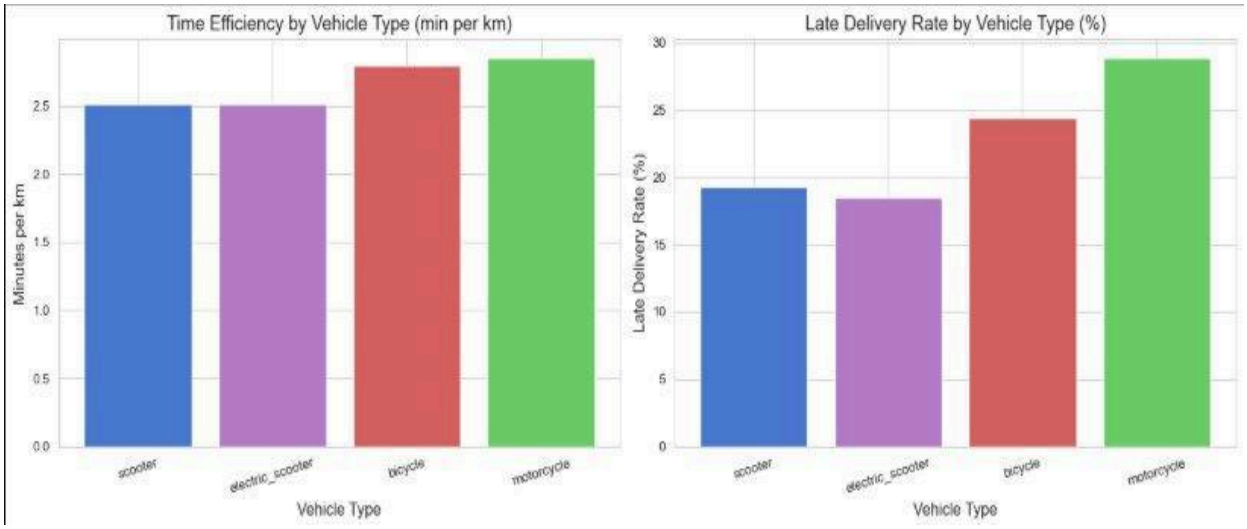

Figure 3. Left: Minutes per km by vehicle type confirming motorcycles as least efficient. Right: Late delivery rate by vehicle type with motorcycle at 28.8% versus 18-19% for scooters.

Liquidity Interpretation: The motorcycle efficiency deficit of 2.85 minutes per km versus 2.51 for scooters represents a structural drag on marketplace throughput. With motorcycles handling 58% of all orders, this inefficiency is amplified across the majority of the platform's order volume. Shifting 18 percentage points of order volume from motorcycles to scooters would reduce average delivery time by approximately 3 minutes per affected order.

6.4 City-Level Fulfillment Performance

The bubble chart below maps each city's average delivery time against its late delivery rate, with bubble size representing order volume and colour representing average driver rating. The chart reveals a two-tier market structure: a high-performing cluster of large cities in the lower-left quadrant, and a smaller group of underperforming cities isolated in the upper-right.

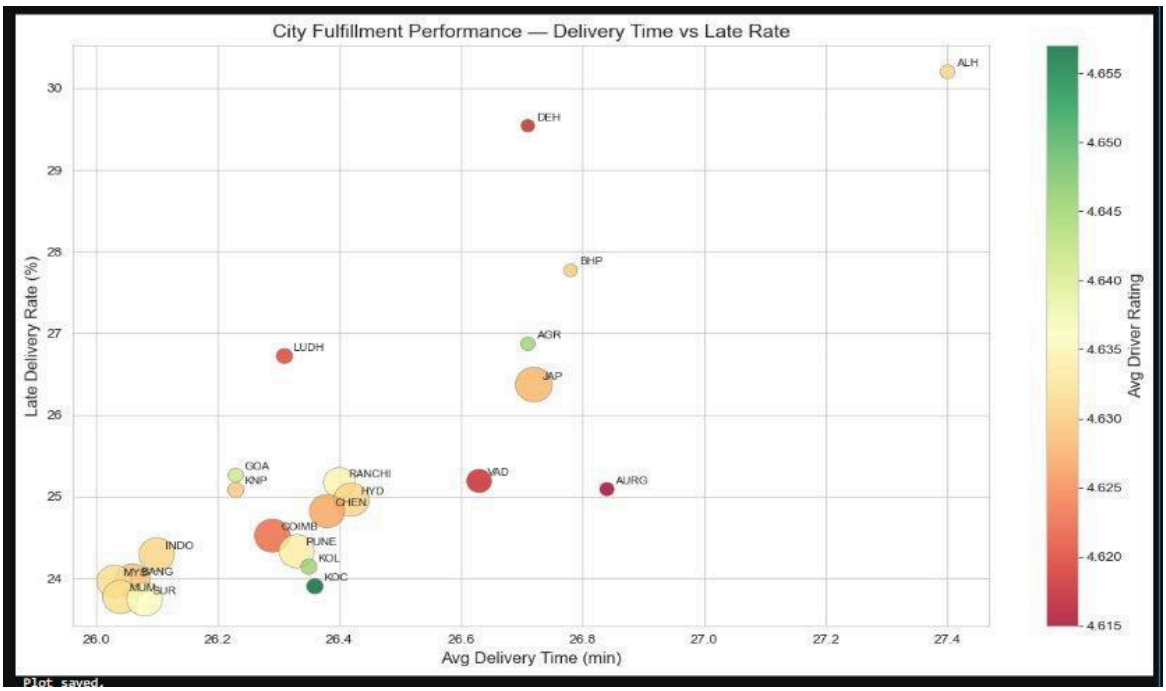

Figure 4. City fulfillment performance map. Bubble size = order volume. Colour = average driver rating (green = higher, red = lower). ALH sits isolated as the worst performer on both dimensions.

Critically, the driver rating colour gradient is near-uniform across all cities, indicating that city-level performance differences are not attributable to driver quality variation between markets. They are more likely driven by local differences in delivery distance distribution and vehicle fleet composition.

7. Retention and Customer Experience Analysis

7.1 Methodology

This dataset does not contain customer identifiers. Direct measurement of repeat ordering behaviour and retention rates is therefore not possible. As an alternative, orders were segmented into four experience cohorts based on delivery time, and these cohorts were used as proxies for customer satisfaction and churn risk. The underlying assumption that slower deliveries correlate with higher churn probability is consistent with published research in the food delivery sector.

7.2 Experience Cohort Analysis

Experience Cohort	Orders	Share	Avg Driver Rating	Avg Distance
Excellent (under 19 min)	11,328	**27.3%**	**4.74**	7.75 km
Good (20-26 min)	11,256	27.1%	4.71	9.24 km
Acceptable (27-32 min)	8,619	20.8%	4.63	9.55 km
Poor (over 32 min)	10,295	**24.8%**	**4.43**	**12.56 km**

Driver rating and average distance both deteriorate monotonically as experience quality worsens. The Poor cohort is characterised by significantly longer delivery distances (12.56 km versus 7.75 km in the Excellent cohort) and materially lower driver ratings (4.43 versus 4.74), confirming that both levers must be addressed simultaneously to move customers from the Poor into the Excellent cohort.

7.3 The Rating Threshold Effect

Granular analysis of delivery time by individual rating value reveals a non-linear relationship with a sharp discontinuity between ratings of 4.4 and 4.5. This is one of the most operationally significant findings in the analysis.

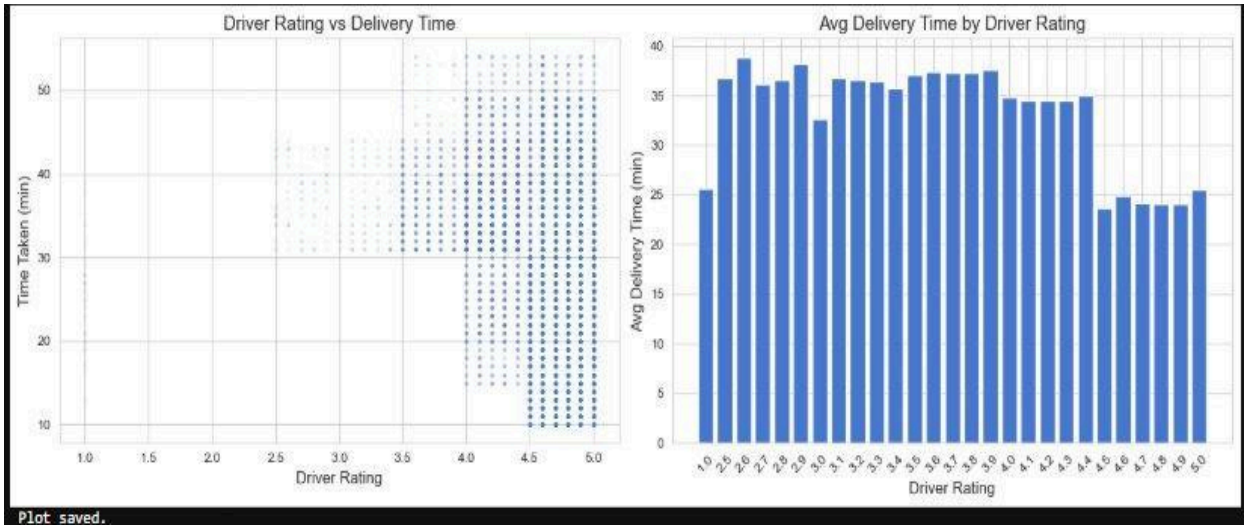

Figure 5. Left: Scatter plot of driver rating vs. delivery time. Right: Average delivery time by binned rating, showing the sharp performance cliff at the 4.5 threshold.

Finding: Drivers rated 4.4 average 34.96 minutes per delivery with a 67% late rate. Drivers rated 4.5 average 23.64 minutes with a 10.6% late rate. This represents a difference of 11.3 minutes and 56 percentage points in late delivery rate within a 0.1-point rating increment. No other variable in the dataset produces a comparably discrete threshold effect.

7.4 Experience Cohort Composition

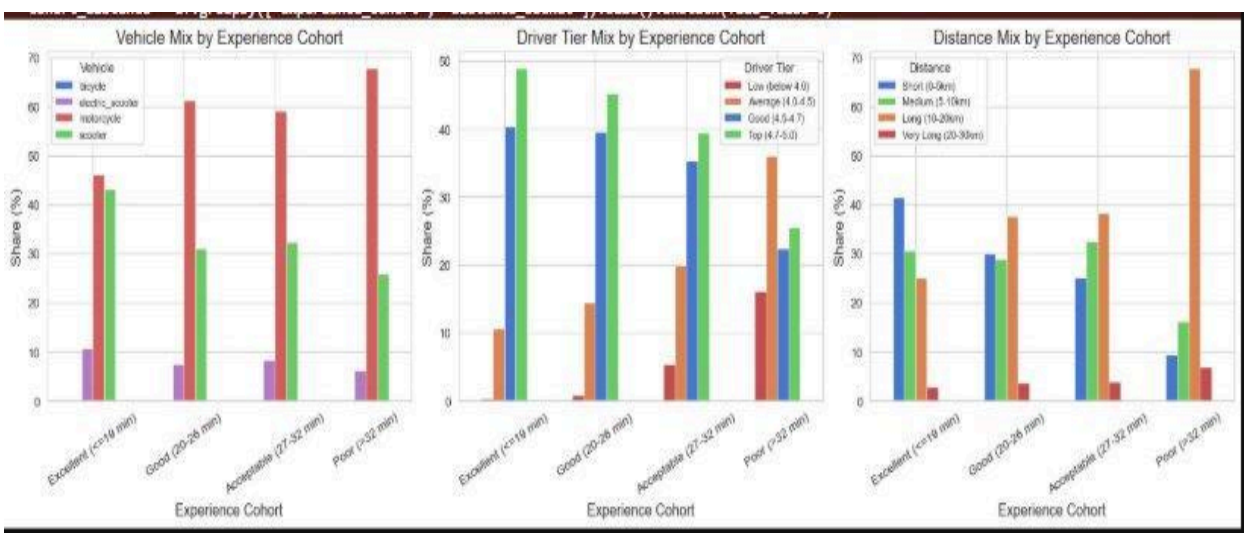

Figure 6. Experience cohort composition by vehicle type (left), driver tier (middle), and distance bucket (right). The Poor cohort is consistently characterised by higher motorcycle share, lower driver tier, and longer routes.

The composition charts confirm that all three primary drivers of delay vehicle type, driver quality, and distance converge in the Poor experience cohort. Operational interventions targeting any one of these variables will produce improvement, but the greatest impact will come from addressing all three in coordination.

8. Operational Efficiency Diagnostics

8.1 Vehicle Type Deep Dive

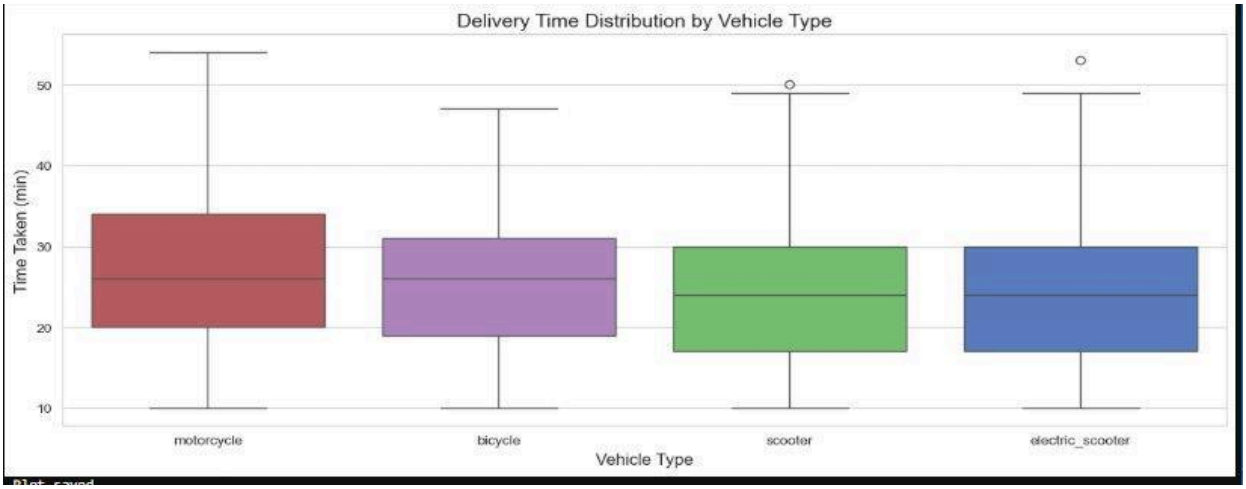

Figure 7. Delivery time distribution by vehicle type. Motorcycle shows a higher median, wider interquartile range, and longer upper whisker relative to scooter and electric scooter.

The interquartile range for motorcycles spans approximately 15 to 34 minutes, compared to approximately 15 to 30 minutes for scooters. This wider spread indicates not only a higher average delivery time but also greater inconsistency a characteristic that is particularly damaging to customer experience because it creates unpredictable wait times.

8.2 Driver Age and Performance

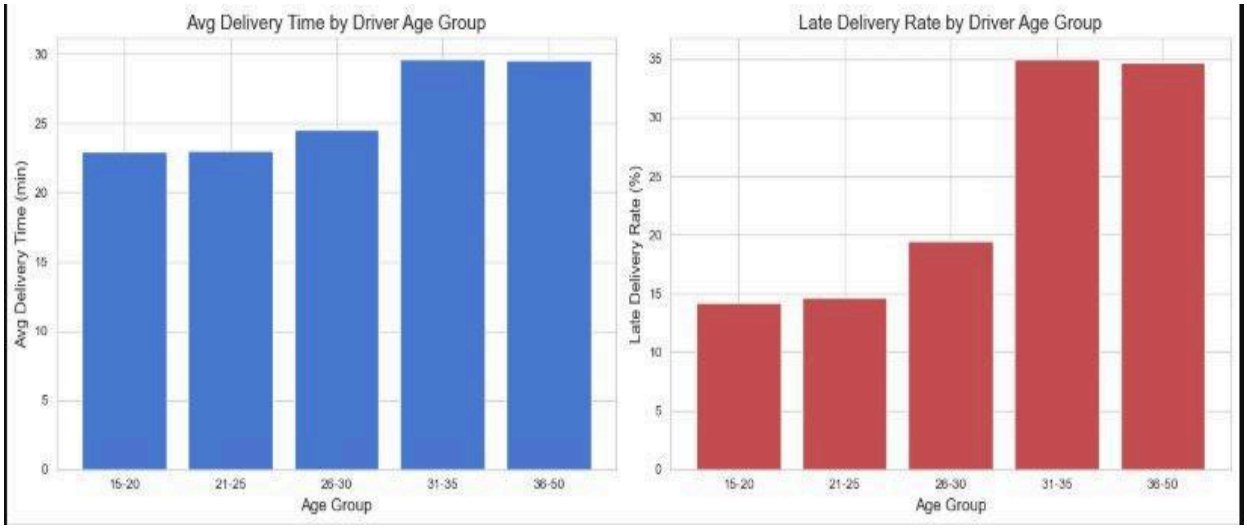

Figure 8. Average delivery time (left) and late delivery rate (right) by driver age group. A performance cliff is visible between the 26-30 and 31-35 age groups.

Drivers aged 15 to 25 deliver in an average of 23 minutes with late rates of 14 to 15 percent. Drivers aged 31 to 50 average approximately 29 to 30 minutes with late rates of 34 to 35 percent. The performance gap of approximately 7 minutes and 20 percentage points cannot be attributed to route distance, which is nearly identical across all age groups. Possible explanatory factors include route familiarity, technological proficiency with navigation applications, and physical agility in urban environments.

Note: Age-based performance differences should be interpreted with care and should not be used as a basis for discriminatory hiring decisions. The more operationally robust intervention is to use the rating-based quality threshold, which is a direct and observable performance measure rather than a demographic proxy.

8.3 Driver Rating Distribution

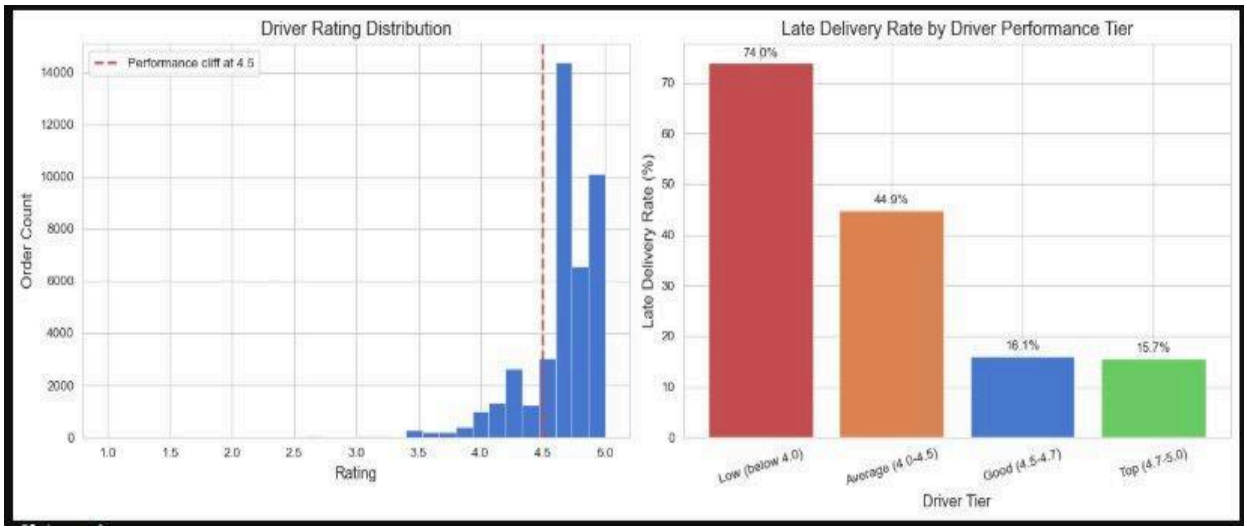

Figure 9. Left: Driver rating distribution showing the majority of drivers cluster above 4.5, with a smaller tail below. Right: Late delivery rate by driver tier quantifying the cliff effect.

The rating distribution is heavily left-skewed toward higher values, which is typical of gig economy platforms where low-rated workers exit the platform over time. Despite this, the 10,491 orders handled by sub-4.5-rated drivers represent a material operational liability, producing late delivery rates that are three to five times higher than those of well-rated drivers.

8.4 Distance Inefficiency by Vehicle Type

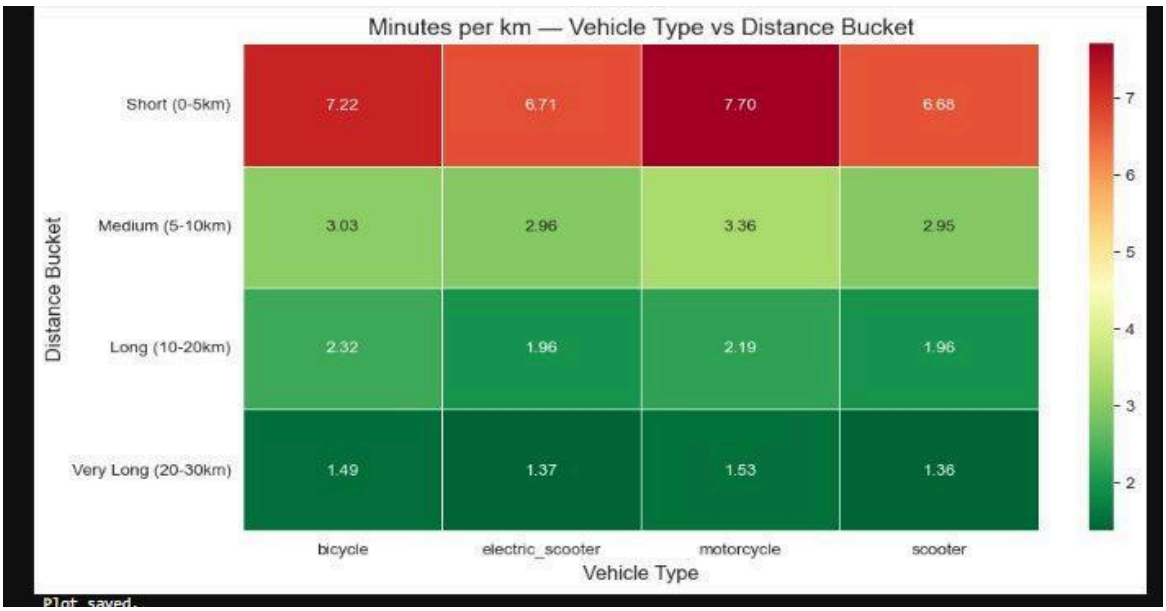

Figure 10. Heatmap of minutes per km by vehicle type and distance bucket. Red indicates higher inefficiency. Motorcycle is consistently the worst performer, particularly on short routes.

An important nuance emerges from this analysis: motorcycle inefficiency is most pronounced on short routes (7.70 minutes per km versus 6.68 for scooters), but the gap narrows substantially on long routes (1.53 versus 1.36 minutes per km). This suggests that motorcycles may be better suited to longer, open routes while scooters are superior for urban short-distance delivery. A distance-aware vehicle routing strategy could leverage this insight to reduce inefficiency without eliminating motorcycles from the fleet entirely.

8.5 City-Level Variance

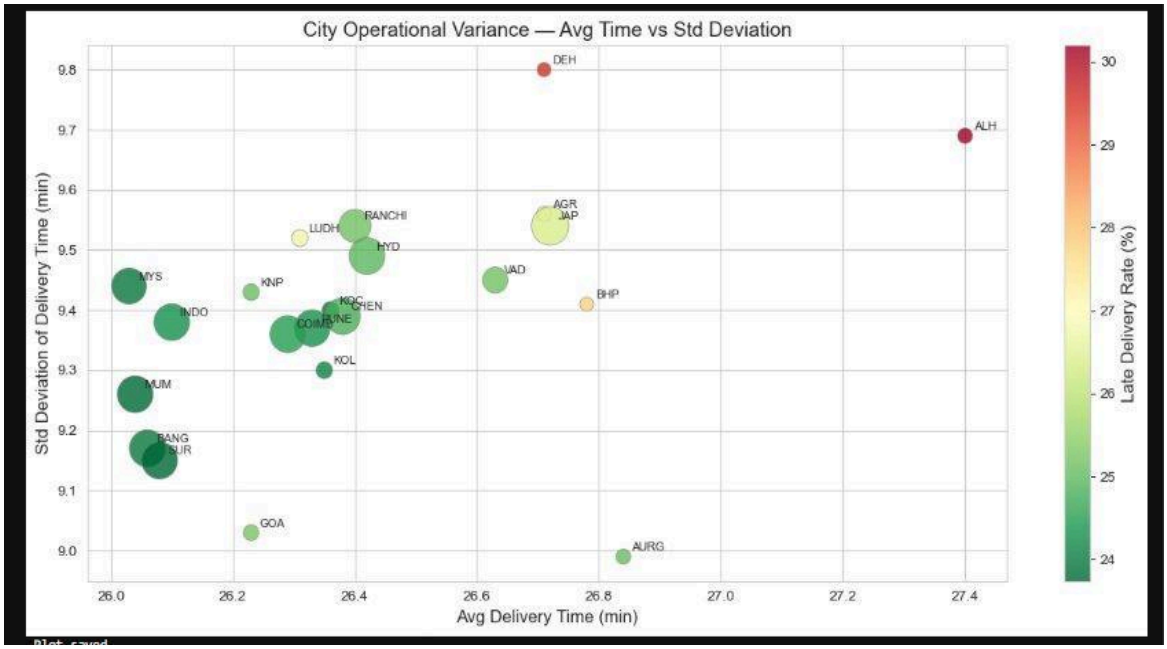

Figure 11. City operational variance average delivery time vs. standard deviation. Cities in the upper-right quadrant (ALH, DEH) exhibit both high average times and high variance, indicating inconsistent operations.

The standard deviation of delivery time ranges from 8.99 minutes (AURG) to 9.80 minutes (DEH). While the range is relatively narrow, cities with higher variance are more likely to produce tail-end slow deliveries that fall into the Poor experience cohort and generate churn risk. BANG and SUR represent the most operationally consistent markets and serve as internal benchmarks for what good looks like.

8.6 Order Type Analysis

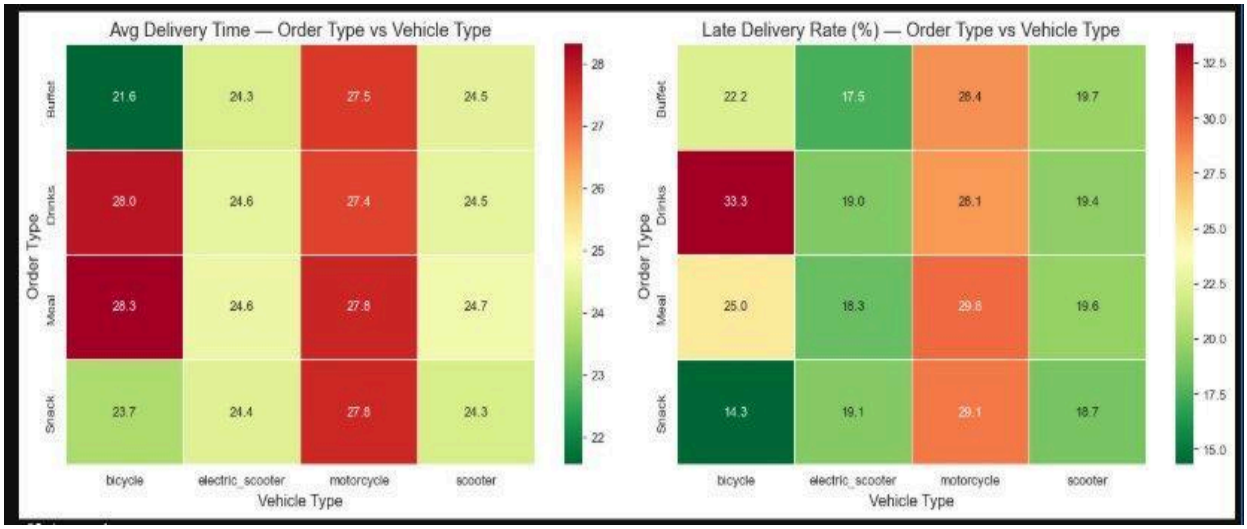

Figure 12. Heatmap of average delivery time (left) and late delivery rate (right) by order type and vehicle type. Motorcycle underperformance is consistent across all four order categories.

Order type has a negligible effect on delivery performance. Average delivery times span only 0.31 minutes across all four categories (Meal: 26.51 min, Drinks: 26.20 min). More importantly, the motorcycle underperformance pattern is perfectly consistent regardless of what is being delivered. This rules out order complexity as an explanatory variable and further isolates vehicle type and driver quality as the primary operational levers.

9. Predictive Modeling

9.1 Problem Framing

Two predictive tasks were defined to quantify the relationship between available features and delivery outcomes:

- Regression: predict continuous delivery time in minutes using driver characteristics, distance, vehicle type, order type, and city.
- Classification: predict whether an order will be late (binary flag) using the same feature set.

All categorical variables were label-encoded. The dataset was split 80/20 into training (33,198 orders) and test (8,300 orders) sets using a fixed random seed for reproducibility. Class balance was preserved across the split, with a late delivery rate of 24.9% in training and 24.3% in test.

9.2 Model Performance

Model	Task	Primary Metric	Secondary Metric
Linear Regression	Delivery Time Prediction	RMSE: 7.961 min	R2: 0.277
Random Forest Regressor	Delivery Time Prediction	**RMSE: 7.696 min**	**R2: 0.324**
Logistic Regression	Late Delivery Flag	Accuracy: 81.2%	F1 (Late): 0.50
Random Forest Classifier	Late Delivery Flag	**Accuracy: 82.1%**	**F1 (Late): 0.59**

9.3 Feature Importance

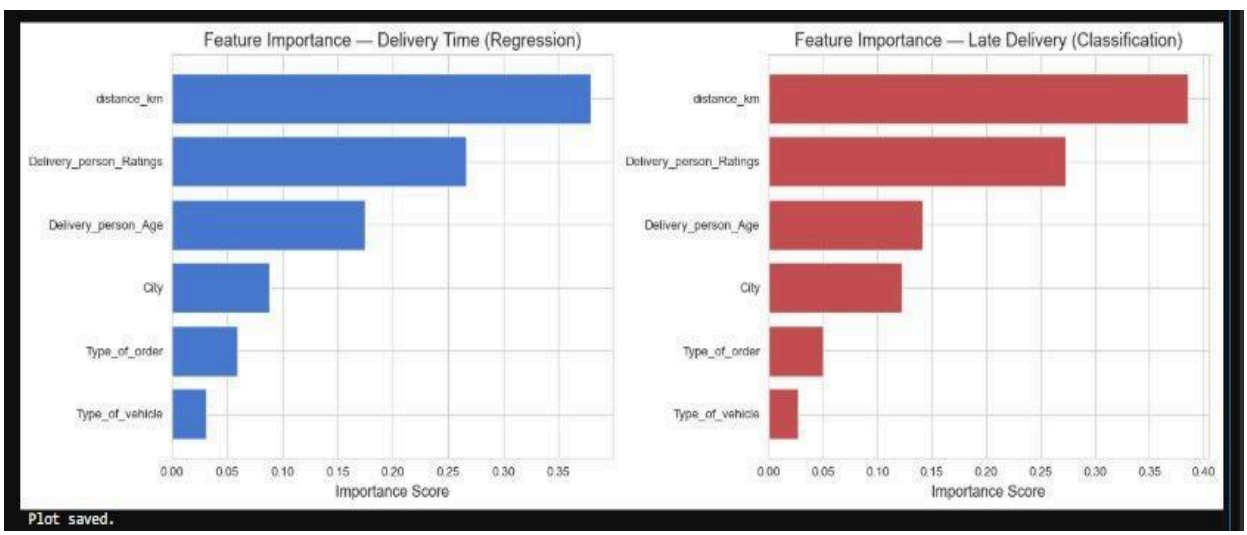

Figure 13. Feature importance from Random Forest models for delivery time regression (left) and late delivery classification (right). Both models agree on the same feature ranking.

Both models produce an identical feature importance ranking, providing strong convergent evidence for the hierarchy of operational levers:

Rank	Feature	Importance	Business Interpretation
1	**Distance (km)**	**38-39%**	The single strongest driver of delay. Long routes are structurally high-risk.
2	**Driver Rating**	**27%**	The most actionable lever. A 4.5 floor would move 10,491 orders to lower-risk drivers.
3	**Driver Age**	**14-17%**	Younger drivers outperform significantly. Proxies for routing proficiency and agility.
4	**City**	**9-12%**	Geographic variation is real but secondary to supply-side factors.
5	**Order Type**	**5-6%**	Minimal impact. Order category is not a meaningful operational variable.
6	**Vehicle Type**	**3%**	Low standalone importance effect mediated through driver rating and age correlations.

Modeling Note: The R2 of 0.32 indicates that approximately 68% of delivery time variance is unexplained by the available features. This is expected given the absence of timestamp data, weather conditions, traffic information, and restaurant preparation time. The models are sufficient for directional feature ranking and operational flagging, but should not be used for precise delivery time prediction in a production setting without enriched data.

10. Strategic Recommendations

10.1 Current State Summary

Metric	Value	Context
Platform-wide late delivery rate	**24.8%**	1 in 4 orders exceeds 32 minutes
Monthly revenue at risk	**$61,770**	Conservative proxy estimate
Orders handled by sub-4.5 drivers	**10,491**	25.3% of all orders highest-risk cohort
Motorcycle order share	**58.3%**	Fleet majority using least efficient vehicle
Long-route late rate (over 10km)	**40.2%**	Affects 46% of all orders
Worst-performing city (ALH)	**30.2%**	6.4 percentage points above best performer

10.2 Priority 1 Enforce a Driver Quality Floor

Recommendation: Establish 4.5 as the minimum active deployment rating for all delivery drivers. Drivers falling below this threshold should be placed in a structured performance improvement program before returning to live fulfillment.

Rationale: The 4.5 rating threshold is the single sharpest performance discontinuity identified in the dataset. It is not a gradual improvement it is a binary threshold. Drivers above 4.5 deliver in approximately 24 minutes with a 16% late rate. Drivers below 4.5 deliver in 30 to 36 minutes with late rates of 45 to 74 percent. This pattern is consistent across all vehicle types and cities, meaning it is a driver quality effect rather than a confounded variable.

The 4.5 threshold also aligns naturally with the rating distribution; the majority of the platform's drivers already meet this standard, meaning the policy change would affect a minority of supply-side participants while disproportionately improving outcomes for customers.

Estimated impact: The 10,491 orders currently handled by sub-4.5-rated drivers experience an average late rate of approximately 57 percent. If these orders were reassigned to Good-tier drivers (16% late rate), approximately 4,301 late deliveries would be prevented per dataset cycle.

10.3 Priority 2 Rebalance the Fleet Toward Scooter

Recommendation: Reduce motorcycle order share from 58% to a target of 40% over two quarters through scooter and electric scooter onboarding incentives and distance-aware order routing.

Rationale: Motorcycles operate at 2.85 minutes per kilometre versus 2.51 for scooters a 13.5% efficiency deficit at comparable distances. This gap is not attributable to the routes assigned to motorcycles, as average distances are nearly identical across vehicle types. It is a vehicle-level efficiency characteristic that manifests at scale.

Importantly, the heatmap analysis reveals that motorcycle inefficiency is most severe on short urban routes, while the gap narrows on longer routes. This suggests a targeted routing strategy: assign scooters and electric scooters to orders under 10 kilometres, where they hold the greatest efficiency advantage, and allow motorcycles to handle longer routes where the performance differential is smaller.

Estimated impact: Shifting 18 percentage points of motorcycle order volume to scooters (approximately 7,469 orders) at the scooter late rate of 19.3% versus motorcycle's 28.8% would reduce late deliveries by approximately 709 orders per dataset cycle.

10.4 Priority 3 Introduce a Distance-Based Routing Cap

Recommendation: Implement a standard delivery routing cap of 10 kilometres. Orders with delivery distances exceeding this threshold should trigger dynamic pricing, premium driver assignment, or restaurant zone reallocation to reduce effective delivery distance.

Rationale: Distance is the strongest statistical predictor of delivery delay at 39% feature importance. Late delivery rates jump from 8.6% on short routes to 40.2% on routes over 10 kilometres. This is not a marginal increase it represents a fundamental change in fulfillment risk profile at the 10-kilometre mark. The fact that rates do not continue to increase proportionally on Very Long routes (39.7% versus 40.2% on Long routes) indicates that the primary issue is crossing the 10-kilometre threshold, not total distance.

In practice, reducing average delivery distance requires a supply-side strategy as well as a routing strategy. Increasing restaurant partner density in underserved delivery zones would reduce the distance between order origins and destinations structurally, rather than relying purely on algorithmic routing limits.

Estimated impact: Long and Very Long routes account for 46% of all orders and generate the majority of late deliveries. Reducing their late rate from 40% to the Medium route benchmark of 14.8% through a combination of routing caps and restaurant density improvements could prevent approximately 4,848 late deliveries per dataset cycle.

10.5 Priority 4 Targeted City-Level Intervention

Recommendation: Launch targeted supply-side investment programs in ALH, DEH, and BHP, the three cities forming the high-risk late-delivery cluster. Interventions should include driver recruitment campaigns, vehicle subsidy programs, and restaurant partner density reviews.

Rationale: These three cities consistently rank at the bottom of the city performance table across average delivery time, late delivery rate, and delivery time variance. ALH has a 30.2% late rate 6.4 percentage points above the best-performing city, SUR. Critically, average driver ratings are near-uniform across all cities, confirming that the underperformance in ALH, DEH, and BHP is not attributable to driver quality differences but rather to structural factors such as delivery distance distribution and fleet composition.

A focused program to improve the vehicle mix and reduce average delivery distances in these markets, benchmarked against the operational profile of BANG and SUR, would likely close a significant portion of the performance gap.

10.6 Quantified KPI Impact Projections

Intervention	Late Orders Prevented	Mechanism
Driver quality floor (4.5 minimum)	**4,301**	Sub-4.5 orders reassigned to Good-tier drivers
Fleet rebalancing (motorcycles to 40%)	**709**	18% of motorcycle orders shifted to scooter fleet
Distance routing cap (10 km max)	**4,848**	Long/Very Long routes reduced to Medium route late rate
Combined (gross, before overlap)	**9,858**	Theoretical maximum assuming no overlap
Combined (adjusted, 50% overlap discount)	**4,929**	Conservative estimate accounting for co-occurring factors

Applying a 50% overlap discount accounts for the reality that many late orders are affected by multiple factors simultaneously: a long route assigned to a motorcycle driven by a low-rated driver, for example. After this adjustment, the projected outcome is:

Metric	Value	Context
Current late delivery rate	**24.8%**	Platform-wide baseline
Projected late delivery rate	**12.9%**	After coordinated three-intervention program
Estimated customers retained	**985**	At 20% churn probability per late order
Monthly recurring revenue recovered	**$29,550**	At $15 order value, 2 orders per customer per month

10.7 Metrics to Track Going Forward

KPI Category	Metric and Target
Primary KPI	Late delivery rate target below 15% within two quarters
Primary KPI	Average delivery time per city target below 25 minutes
Supply KPI	Driver rating distribution target 90%+ of active drivers above 4.5
Supply KPI	Fleet mix target scooter + electric scooter share above 50%
Logistics KPI	Average delivery distance per order target below 10 km
City KPI	Late rate in ALH, DEH, BHP weekly monitoring with monthly review
Retention KPI	Repeat order rate by experience cohort (requires customer ID integration)
Revenue KPI	Monthly revenue at risk from late deliveries track against $61,770 baseline

11. Limitations and Next Steps

11.1 Analytical Limitations

The following constraints limit the precision and scope of conclusions drawn in this report. They are documented to enable informed interpretation of findings and to guide data collection priorities for future analysis.

1. No timestamps available. The absence of order timestamps prevents analysis of peak-hour effects, day-of-week patterns, and seasonal demand variation. These factors are likely responsible for a significant portion of the 68% unexplained variance in the predictive models.
2. No customer identifiers. True measurement of customer retention, repeat order rates, and churn probability is not possible with this dataset. All retention-related figures in this report are directional proxies based on industry benchmarks and should be treated as order-of-magnitude estimates.
3. No revenue or order value data. All financial impact calculations use a $15 average order value proxy. Actual impact will vary by market, order type, and time period.
4. No restaurant preparation time. The Time_taken(min) outcome variable conflates restaurant preparation time with delivery transit time. Isolating transit time from preparation time would enable more precise operational diagnostics and would likely improve model performance.
5. No weather or traffic data. External environmental factors are entirely absent from the dataset. Their inclusion would substantially improve the R2 of the predictive models and enable more precise causal attribution of delays.
6. Haversine distance is a straight-line proxy. Road distances in dense urban environments can exceed straight-line estimates by 20 to 40 percent. All distance-related conclusions should be validated against actual routing data before being used as the basis for operational decisions.

11.2 Next Steps

The following actions are recommended to extend and operationalise the findings of this analysis:

7. Enrich the dataset with order timestamps to enable time-of-day and day-of-week segmentation. This is the highest-priority data gap.
8. Integrate customer identifiers to measure true retention impact of late deliveries and validate the 20% churn assumption against observed repeat ordering behaviour.
9. Separate restaurant preparation time from delivery transit time to enable targeted operational intervention at the restaurant versus driver level.
10. Incorporate traffic and weather API data to build a production-grade late delivery prediction model with materially higher explanatory power.
11. Run a controlled A/B test of the 4.5 driver rating minimum in a single city before platform-wide rollout, measuring late delivery rate, driver supply, and customer satisfaction scores.
12. Build a real-time late delivery risk score using the Random Forest classifier to trigger proactive customer communications when delivery risk is elevated reducing the satisfaction impact of unavoidable delays.
13. Commission a restaurant density audit in ALH, DEH, and BHP to determine whether increasing partner coverage in underserved delivery zones would structurally reduce average delivery distances in these markets.